

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

NAME: Naveen G
REGNO: 192411019

COURSE OUTCOME COVERED

CO -2 : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool : Design Task

Weightage: 20%

QUESTIONS:

Total Marks: 20

Question 1: Responsive Navigation Bar Design - Marks(4)

Suppose that an e-commerce website requires a responsive navigation bar that adapts across desktop, tablet, and mobile devices.

- (a) Identify key components required in a navigation bar for an e-commerce website.
- (b) Design a responsive navigation bar using appropriate CSS techniques.
- (c) Demonstrate how the navigation adapts to smaller screens (e.g., hamburger menu).
- (d) Evaluate the usability and suggest improvements for better user experience.

ShopEase

HomeProductsCategoriesOffersContact

Search products

0 Cart

Login

(a) Key Components

- Brand Logo
- Navigation Links
- Search Bar
- Shopping Cart
- Login Button
- Hamburger Menu

(b) CSS Techniques

- Flexbox Layout
- Media Queries
- Gradient Background
- Hover Effects
- Sticky Header
- Transitions

(c) Responsive Design

- Desktop Full Menu
- Tablet Compact Layout
- Mobile Hamburger Menu
- Touch Friendly Buttons
- Smooth Menu Animation

(d) Evaluation

- Easy Navigation
- Fast Access
- Accessible Layout
- Responsive on All Devices
- Modern User Experience

Live Evaluation

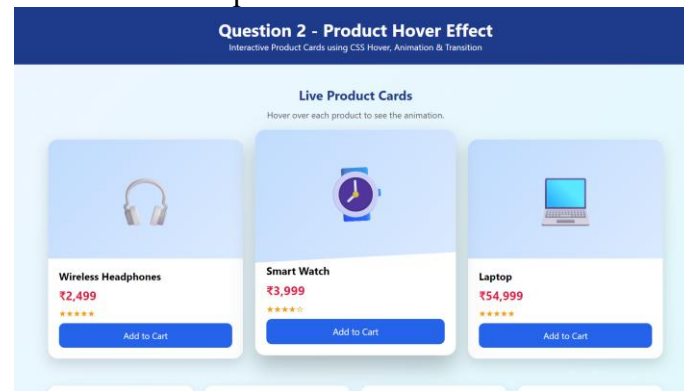
The responsive navigation bar adapts to different screen sizes. Resize the browser window to observe the hamburger menu on mobile devices.

Device	Navigation Style	User Experience
Desktop	Horizontal Menu	Excellent
Tablet	Compact Menu	Very Good
Mobile	Hamburger Menu	Excellent

Question 2: Product Hover Effect - Marks(4)

Suppose that an e-commerce website wants to enhance user interaction through visual effects on product items.

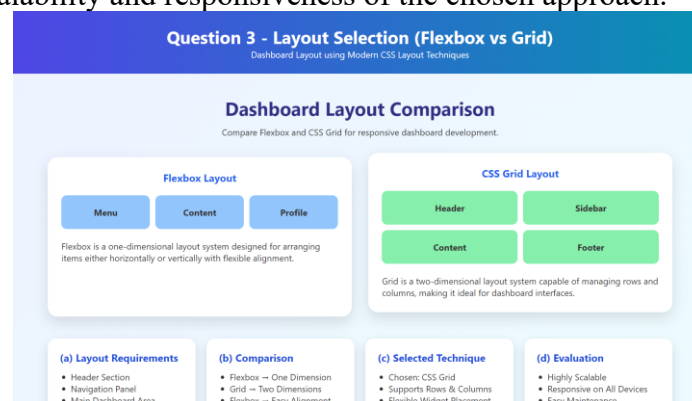
- Identify suitable CSS properties for hover effects.
- Suggest appropriate **CSS animation or transition** for product interaction.
- Design a hover effect for a product card.
- Evaluate performance and user experience considerations.



Question 3: Layout Selection (Flexbox vs Grid) - Marks(4)

Suppose that a dashboard layout needs to be designed for a web application.

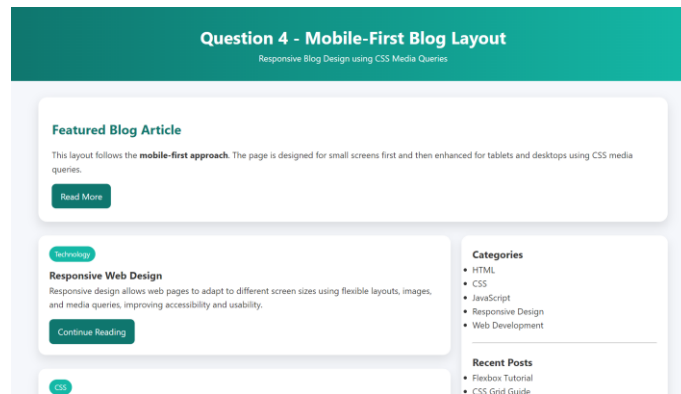
- Identify layout requirements of a dashboard (rows, columns, alignment).
- Compare **Flexbox and Grid** based on layout needs.
- Choose the most suitable layout technique and justify your choice.
- Evaluate the scalability and responsiveness of the chosen approach.



Question 4: Mobile-First Blog Layout - Marks(4)

Suppose that a blog page must be designed following a mobile-first approach.

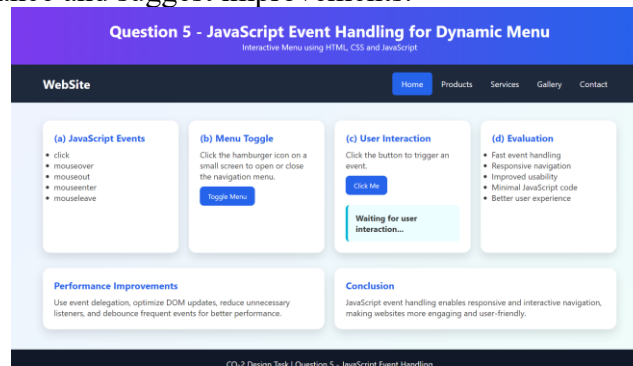
- Identify key sections of a blog layout.
- Design a mobile-first layout using CSS.
- Demonstrate how the layout scales for larger screens using media queries.
- Evaluate accessibility and readability of the design.



Question 5: JavaScript Event Handling for Dynamic Menu - Marks(4)

Suppose that a website includes a dynamic menu that responds to user interactions.

- Identify appropriate JavaScript events for menu interaction.
- Design event handling logic for menu toggle functionality.
- Implement interaction for user actions (click/hover).
- Evaluate performance and suggest improvements.



Code of Conduct:

I / Naveen G / 192411019 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Concept	25	Demonstrates complete understanding of design requirements and concepts	Good understanding with minor gaps	Basic understanding with some errors	Poor understanding or incorrect concepts
Design / Implementation	30	Well-structured, efficient, and responsive design implementation	Correct implementation with minor issues	Basic implementation with inconsistencies	Incorrect or incomplete implementation
Use of Tools (CSS/JS)	20	Effective and advanced use of tools/techniques	Appropriate use of tools	Limited or inconsistent use	Incorrect or minimal use
Analysis & Evaluation	15	Insightful analysis with strong justification and optimization	Good analysis with justification	Basic analysis with limited depth	Poor or no analysis
Documentation / Clarity	10	Clear, well-structured, and properly presented solution	Mostly clear with minor issues	Basic clarity, missing details	Poor or unclear documentation